

The Price of the Early Exit: ncu After GPU Periodicity

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Binary: `binary-v7-gpu-periodicity` (2abc909, on main since 41b6c2e), captured with Nsight Compute 2026.2. Baseline: experiment 16’s tracked capture of `binary-v5-affinity` (fc33e29) — the device `diverge()` is unchanged v5→v6, so that capture is the immediately-pre-check kernel. Raw data: `experiments/24-ncu-gpu-periodicity/`.

Resumen

Reports 16/17 pinned this kernel at exactly 100.0% warp execution efficiency on interior content, and report 21 cited that number to defer the GPU-side periodicity check (“expect a loss”). Report 23 falsified the conclusion at the wall (−20.5% hybrid); this report replaces the stale counters. Same method as experiment 16 (same four content-regime specs, same kernel caps, same metrics; aggregation validated to reproduce experiment 16’s summary exactly; clocks match within 1%, so durations compare). The price of the early exit is small and precisely locatable: interior warp efficiency falls only 100.0 → 96.3% (minimum kernel 95.0%) while interior kernels get 9.9× faster; the isolated overhead — pure-exterior content, where the check never fires — is +12.4%, half the 25–30% estimate, because `nvcc` short-circuits the two anchor compares into one always-executed DSETP. The cost concentrates on boundary content (+23%, kernels production routes to the CPU anyway); the production mix nets 9.4× per kernel. The FP64 pipe stays at ~85% of peak in every regime, before and after: the kernel remains FP64-bound, and every metric that “degraded” — warp efficiency, achieved occupancy — reads worse precisely because the work now ends early.

1. Motivation: replace the number the wrong prediction leaned on

Report 16 measured interior kernels at exactly 100.0% warp execution efficiency and 85.6% of FP64 peak; report 17 reproduced both on the four zoom-point geometries. Those were correct measurements read as health: report 21’s recommendation 6 cited them to predict the GPU-side periodicity check would lose. Report 23 measured the check instead — mix-controlled kernel 2.34×, hybrid 29.83 → 23.72 s, output byte-identical — and identified the misreading: on interior content, “100% efficiency, zero early exits” is the waste (every lane of every warp grinding provably-useless iterations to `MAXITER` in perfect lockstep). What report 23 could not say is what the counters look like now: how much warp efficiency the early exit actually spends, where the two-compares-per-iteration overhead lands, and whether the FP64 roofline moved. Without a measured replacement, the 100.0% baseline stays on record as a target someone could “restore” by deleting the check. This experiment re-baselines all of it.

2. Methodology

Identical to experiment 16, on the merged v7 binary: the same four content-regime specs reused verbatim (**interior** = main cardioid, **maxIter** 2000–3000; **exterior** = fast escape; **boundary** = seahorse valley; **canonical** = the production zoom), GPU-only runs (1 1 0.1 32768 1 0 — the single GPU thread sees every region class), **—set full** on 6/6/60/60 kernels plus a cheap warp/branch **—metrics** pass on 6/6/300/300. Counters are admin-gated; capture ran via **sudo capture.sh**, analysis unprivileged via **analyze.py**.

Baseline. Experiment 16’s tracked **analysis/summary.csv**. The device **diverge()** is unchanged between **binary-v5-affinity** (which experiment 16 captured) and v6 (a CPU-only change), so no re-capture of the old kernel is needed. **analyze.py** reproduces experiment 16’s aggregation exactly — validated in-session against its tracked summary on all four regimes (simple mean over captured kernels of the same raw-page columns). One slip found in the reference: experiment 16’s **exterior dur_ms** is recorded in μs (614.869); corrected to 0.615 ms here.

Comparability. Warp/branch efficiency, pipe and SOL percentages, and occupancy are clock-independent ratios. Durations normally carry a thermal caveat across sessions, but the recorded GPC clocks match within 1% (v6: 1.317–1.345 GHz; v7: 1.322–1.342 GHz), so the duration ratios are quoted as measured. The decomposition is deterministic and identical in both binaries (report 23), so capture N sees the same N regions.

3. Results

Tabla 1: Full-set captures, v6 baseline (experiment 16) vs v7 (this capture); each cell is v6 $\rightarrow$ v7. Read row-wise: the check’s entire metric cost is the interior warp-efficiency drop, the +12–23% on content that never cycles, and 6 registers; the FP64 pipe does not move.

metric	interior	exterior	boundary	canonical
warp exec. eff. (%)	100.0 $\rightarrow$ 96.3	100.0 $\rightarrow$ 100.0	92.6 $\rightarrow$ 92.4	94.4 $\rightarrow$ 91.9
branch eff. (%)	100.0 $\rightarrow$ 99.7	100.0 $\rightarrow$ 100.0	99.6 $\rightarrow$ 99.8	96.4 $\rightarrow$ 97.8
FP64 pipe (% peak)	85.6 $\rightarrow$ 85.3	85.6 $\rightarrow$ 84.9	85.3 $\rightarrow$ 85.1	84.1 $\rightarrow$ 84.0
compute SOL (%)	85.5 $\rightarrow$ 85.1	85.4 $\rightarrow$ 84.6	78.1 $\rightarrow$ 77.9	69.4 $\rightarrow$ 66.6
achieved occup. (%)	98.9 $\rightarrow$ 89.5	94.0 $\rightarrow$ 91.5	84.3 $\rightarrow$ 84.6	79.1 $\rightarrow$ 77.4
mean kernel (ms)	726.4 $\rightarrow$ 73.5	0.615 $\rightarrow$ 0.691	16.8 $\rightarrow$ 20.7	30.7 $\rightarrow$ 3.25
duration ratio	9.9 $\times$	+12.4%	+23.3%	9.4 $\times$
registers/thread	22 $\rightarrow$ 28	22 $\rightarrow$ 28	22 $\rightarrow$ 28	22 $\rightarrow$ 28

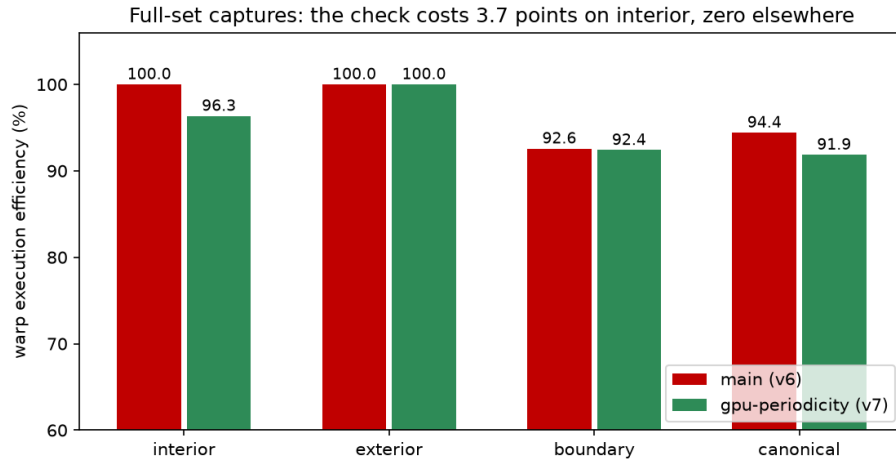


Figure 1: Full-set warp execution efficiency by regime. The check’s entire efficiency cost on the full-set captures is the 3.7-point interior drop (and 2.6 on the production mix); exterior — where no orbit ever cycles — is untouched at 100.0%, and boundary’s structural floor-leaf divergence (report 16) is unchanged.

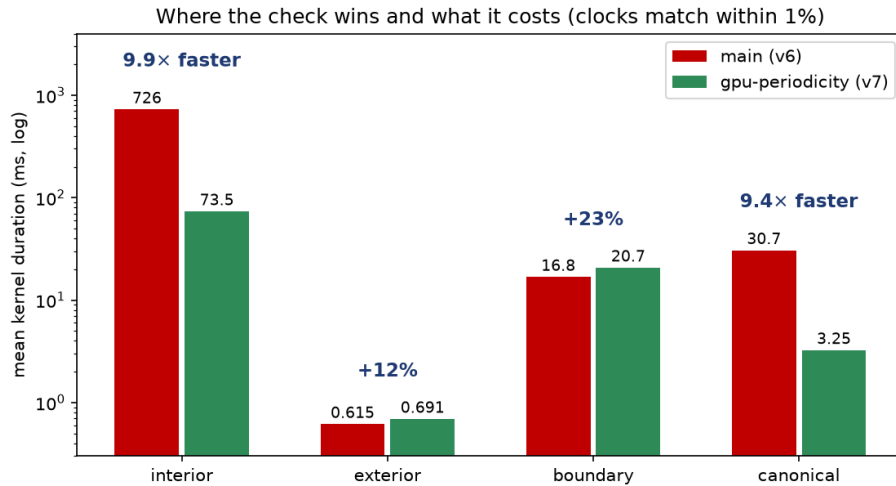


Figure 2: Mean kernel duration per regime, log scale (GPC clocks match within 1% between the sessions). The win lands where the cycles are (interior 9.9×, production mix 9.4×); the cost lands where they aren’t (exterior +12.4%, boundary +23.3%). Production affinity routes boundary floor leaves to the CPU pool, so the +23% barely prices into the production config that report 23 measured at −20.5% wall.

Tabla 2: Per-kernel warp-efficiency distribution from the large `_warp` captures (both binaries reduced identically). Boundary is statistically unchanged — the floor-leaf divergence is a property of region size, not of the check — while canonical’s distribution shifts down: its big interior kernels now contain the early exits, by design.

regime	kernels	mean (%)	min (%)	below 70%
boundary v6	276	70.7	28.6	47.1%
boundary v7	276	70.2	28.5	47.5%
canonical v6	300	87.6	22.2	12.7%
canonical v7	300	78.1	18.7	32.0%

Divergence structure: boundary unchanged; canonical’s interior kernels absorb the early exits

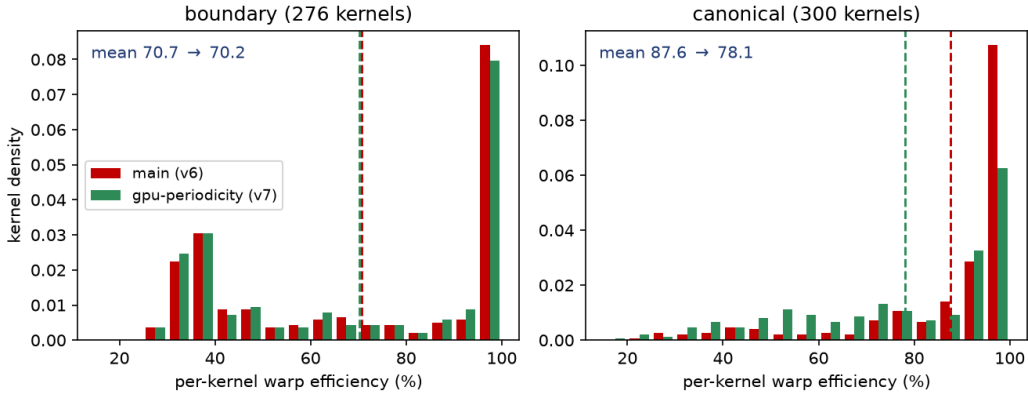


Figure 3: The same distributions as Table 2. Left: boundary histograms lie on top of each other — the check neither adds to nor removes the structural divergence of the 240×135 floor leaves. Right: canonical’s mass moves out of the 95–100% bin into the 40–90% range: the interior kernels that used to run in wasteful lockstep now retire lanes as they detect their cycles.

4. Analysis

4.1. Interior pays 3.7 points because detection is synchronized within a warp

The feared divergence catastrophe is a 3.7-point drop: interior kernels average 96.3% warp efficiency (minimum 95.0%, nothing below 90% in any capture) while running $9.9\times$ faster. The reason detection is so coherent: a warp’s 32 lanes are contiguous pixels of the same interior region converging to the same attracting cycle, and every lane refreshes its Brent anchor at the same iteration numbers (8, 16, 32, ... — the schedule is iteration-indexed, not state-indexed). Neighbouring orbits therefore satisfy the exact-revisit test at nearly the same iteration, and the masked-idle tail between the first and last detection in a warp is short relative to even the shortened kernel. The duration ratio quantifies the effective latency: $9.9\times$ at `maxIter` 2000–3000

and +12.4% per-iteration overhead implies the slowest lane detects at ~ 225 iterations on main-cardioid content — between report 21’s 67 (period-3 bulb) and ~ 900 (deep minibrot), as expected for the period-1 cardioid interior with a slow transient.

4.2. The isolated overhead is +12.4%, not 25–30%: nvcc short-circuits the compare pair

Exterior content prices the check’s overhead with zero benefit mixed in: no escaping orbit ever revisits a state, so the branch never fires, and the regime stays at exactly 100.0% warp efficiency — the +12.4% duration is pure per-iteration cost. Reports 21/23 estimated 25–30% by counting two FP64 compares against a ~ 7 -instruction recurrence; the SASS shows why the measured cost is half that: the compiler compiles `vx == sx && vy == sy` as one always-executed `DSETP.NEU` on `vx`, with the `vy` compare (`DSETP.EQ`) predicated behind it — it almost never issues on non-cycling orbits. One extra FP64-pipe instruction on a loop body of ~ 8 predicts +12.5%; measured, +12.4%. Consistently, the FP64 pipe stays at $\sim 85\%$ of peak in every regime: the check did not move the roofline, it added one slot to the iteration.

4.3. Boundary pays the most (+23%) and produces nothing — and production doesn’t care

Boundary kernels slow $16.8 \rightarrow 20.7$ ms with a warp-efficiency distribution that is statistically identical (mean $70.7 \rightarrow 70.2$, 47% below 70% in both, minimum $\sim 28.5\%$): seahorse-valley content iterates long but either escapes or fails to exactly cycle within the anchor schedule, so it pays the compare on every iteration and the early exit almost never fires. The +23% exceeding the +12.5% static floor is the one number these counters do not fully attribute (candidates: the predicated second compare firing on near-cyclic orbits, and thinner warps hiding the recurrence latency less well); it is also the least consequential — report 16 showed these are the 240×135 pixel-floor leaves, and in the production hybrid the affinity queue routes them to the CPU pool. The production mix (canonical) nets $9.4\times$ per kernel regardless.

4.4. The occupancy drop is the early exit’s signature, not register pressure

Registers rise $22 \rightarrow 28$ (the two anchor doubles and the schedule counter live across the loop), which moves the register occupancy limit from 10 to 8 blocks/SM — both far above the 4-block warp limit that actually binds ($4 \times 256 = 1024$ threads/SM), so theoretical occupancy is unchanged. The achieved drop ($98.9 \rightarrow 89.5\%$ on interior) is the same phenomenon as the warp-efficiency drop seen one level up: warps retire as their lanes detect cycles, blocks drain, and the SM spends a larger fraction of a much shorter kernel partially full. Like warp efficiency, achieved occupancy now reads “worse” because the kernel stops doing useless work.

4.5. The counters reconcile with report 23’s $2.34\times$, and the metrics’ role changes

Report 23 measured the mix-controlled kernel total at $2.34\times$ over all 5,608 regions; these captures show interior-type content at $\sim 9.9\times$ and never-cycling content at

$\sim 1.2\times$ slower. The blend $s/9.9 + (1 - s) \times 1.2 = 1/2.34$ solves to $s \approx 0.70$: seventy percent of the v6 kernel time was interior grind — the GPU-side echo of the leaf-purity census that motivated report 21. The practical conclusion for the metric suite: on this kernel, warp execution efficiency and achieved occupancy are no longer health indicators; they are the accounting of eliminated work. The FP64 pipe percentage ($\sim 85\%$, invariant) remains the real health line — it says the kernel is still FP64-bound, so the next kernel-side lever (if one is ever needed) is still arithmetic, not scheduling.

5. Recommendations

1. Adopt these numbers as the healthy baselines: interior 96.3% warp / 99.7% branch efficiency, canonical distribution mean 78.1% with 32% of kernels below 70%, achieved occupancy 89.5%, FP64 pipe 85%. Reports 16/17’s 100.0% figures are superseded by design; a future capture that reads 100.0% on interior content means the check was removed and $\sim 10\times$ kernel time came back.
2. Do not pursue k -strided compare amortization. The motivation (reports 21/23’s 25–30% overhead estimate) is gone: the short-circuited compare costs +12.4%, and striding the anchor compare to every k -th iteration would delay detection by up to $k - 1$ iterations on a ~ 225 -iteration prize to recover part of one instruction in eight. The measured ceiling is not worth the exactness-argument complication.
3. Proceed to the `scanLine` commit path (report 23, recommendation 2). The kernel side is done: $9.4\times$ on the production mix with the pipe still 85% saturated leaves the GPU thread’s host side (12.8 s vs 10.7 s kernel at steady state) as the only lane cost that has not been attacked.
4. Port the check to `oclkernel.cpp` before the third iGPU repricing (report 23, recommendation 4) — unchanged, with one addition from this data: expect the iGPU’s kernel-side gain to be content-identical ($\sim 10\times$ on interior) but its warp (subgroup) metrics to shift the same way; budget the same identity-gate-against-itself validation.

6. Conclusion

The GPU-side periodicity check’s microarchitectural price is measured and small: interior warp execution efficiency $100.0 \rightarrow 96.3\%$ (minimum 95.0%), achieved occupancy $98.9 \rightarrow 89.5\%$, six extra registers with the theoretical occupancy limit untouched, and +12.4% per-iteration overhead where no cycle exists — half the pre-measurement estimate, because `nvcc` short-circuits the anchor compares. The prize the metrics paid for: interior kernels $9.9\times$, the production mix $9.4\times$ per kernel, reconciling to report 23’s mix-controlled $2.34\times$ via a 70/30 interior/other time split. Boundary floor leaves pay +23% and remain exactly as divergent as report 16 found them — structural, size-bound, and routed to the CPU in production. The FP64 pipe stays at $\sim 85\%$ of peak in every regime: the kernel is still FP64-bound, the roofline did not move, and the metrics that “got worse” are the ledger of the work that no

longer happens. Reports 16/17's 100.0% baselines are hereby replaced; the queued levers are unchanged: **scanLine** host-side commit, then the iGPU port and repricing.